

Sourabh Yadav

Research Scholar

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Research Interests

My research interests lie at the intersection of Large Language Models, Reinforcement Learning, Explainable AI, and AI Safety. I focus on developing reliable, interpretable, and computationally efficient intelligent systems for real-world applications. My work explores reasoning and alignment in foundation models, scalable learning algorithms, uncertainty estimation, model explainability, and the deployment of trustworthy AI systems. I am particularly interested in advancing methods that improve the robustness, transparency, and resource efficiency of modern AI technologies.

Academic Qualifications

Year	Degree/Certificate	Institute/Board	CPI/Percentage
<i>Present</i>	PhD	IIT Kanpur, India	8.00/10
2024	Master of Technology, Systems & Control	IIT Hyderabad, India	8.77/10
2022	Bachelor of Technology, Electrical Engineering	Bhilai Institute of Technology, Durg, India	8.70/10
2017	HSC (XII)	CBSE Board, India	92.40 %
2015	SSC (X)	CBSE Board, India	9.4/10

Research Experience

PhD Thesis (*2024-present*)

IIT Kanpur, India

Mentor: Prof. Nishchal K. Verma, Electrical Engineering, Indian Institute of Technology, Kanpur

Developing hallucination detection and mitigation frameworks for Large Language Models, aimed at improving factual accuracy, reliability, and trustworthiness in generative AI systems. The research combines machine learning and explainable AI techniques to support the safe deployment of foundation models in real-world applications.

Master's Thesis (*2022-24*)

IIT Hyderabad, India

Mentor: Prof. Ketan P. Detroja, Electrical Engineering, Indian Institute of Technology, Hyderabad

Designed adaptive PID and classical PI controllers using reinforcement learning (DDPG, TD3, PPO), achieving improved control performance on multivariable, non-minimum phase, and unstable systems, while investigating transfer learning for controller adaptation across process dynamics.

Publications

Journals / IEEE Transactions:

1. S. Yadav and N. K. Verma, "A Hybrid Framework for Hallucination Detection in Large Language Models," in IEEE Transactions on Artificial Intelligence
2. S. Yadav and N. K. Verma, "Explainable AI for Sentiment Classification with Type-1 Fuzzy & Type-2 Fuzzy Models," in IEEE Transactions on Fuzzy Systems

Conferences:

1. S. Yadav and K. P. Detroja, "Reinforcement Learning based Adaptive PID Controller for Non-minimum Phase Unstable Systems with delay," 2025 11th International Conference on Control, Decision and Information Technologies (CoDIT), Split, Croatia, 2025, pp. 2523-2528
2. S. Yadav and N. K. Verma, "Teaching Small LLMs to Reason Using Reinforcement Learning," 2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2026, pp. 1347-1353

Teaching/Work Experience

- **Teaching Assistant (TA)** duty at IIT Kanpur in the following courses: Artificial Intelligence, Machine Learning, Deep Learning and Its Applications, Industrial Automation and Control, Fuzzy Sets, Logic and Systems & Applications.
- **Teaching Assistant (TA)** for the NPTEL course titled Fuzzy Sets, Logic and Systems & Applications, offered by IIT Kanpur as part of the SWAYAM platform.
- **Tutor** for the Artificial Intelligence, Machine Learning, Deep Learning and Its Applications course at IIT Kanpur.

Professional Memberships & Service

Journal Reviewer:

- IEEE Transactions on Artificial Intelligence

Conference Reviewer:

- European Control Conference (ECC)
- Asian Control Conference (AsCC)

References

1. **Prof. Nishchal K. Verma**
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2. **Prof. Ketan P. Detroja**
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